Feature Selection Report

Customer Personality Segmentation | Module 5: Feature Engineering & Data Preprocessing (CPS-M05)

After feature creation and encoding, the working dataset had 48 columns. This report explains what I removed before clustering, what I kept, and why — the goal is a feature set that's informative for customer segmentation without redundant or noisy columns dragging cluster quality down.

1. Columns Removed

Column	Reason for Removal
ID	Unique identifier per customer — carries no behavioral or demographic signal, and including it would activ
Year_Birth	Superseded by the engineered Customer_Age feature, which is more directly interpretable.
Dt_Customer	Superseded by Customer_Tenure_Days, which expresses the same information as a usable numeric dura
Z_CostContact	Constant value across all 2,237 customers (zero variance) — contributes nothing to differentiating segmen
Z_Revenue	Constant value across all 2,237 customers (zero variance) — same issue as above.
Education (text)	Replaced by Education_Encoded, its ordinal numeric version.
Customer_Activity_Level (text)	Replaced by Customer_Activity_Level_Encoded, its ordinal numeric version.
Kidhome, Teenhome	Combined into Total_Children — kept the combined feature since the household-size signal matters more

2. Features Retained & Why They Matter

40 features survived selection (before scaling; the final scaled/encoded dataset has 41 columns plus ID). They fall into eight groups:

Group	Features	Why It's Kept
Demographics	Income, Customer_Age, Education_Encoded	The core drivers of spending capacity and lifestyle — almost every segmentation study starts here.
Household structure	Total_Children, Family_Size, Marital_Status_* (one-hot)	Household composition strongly shapes purchase volume and category mix (families vs. singles buy differently).
Spending behavior	Total_Spending, MntWines, MntFruits, MntMeatProducts, MntFishProducts, MntSweetProducts, MntGoldProds, Avg_Spending_Per_Purchase	Captures both total value and category mix — two customers with equal Total_Spending can have very different profiles.
Purchase channel behavior	Total_Purchases, NumDealsPurchases, NumWebPurchases, NumCatalogPurchases, NumStorePurchases, Preferred_Shopping_Channel_* (one-hot)	Separates in-store shoppers from online/catalog shoppers — directly actionable for channel-specific marketing.
Price sensitivity	Deal_Dependency	Flags price-sensitive vs. full-price segments, a classic and useful segmentation axis.

Group	Features	Why It's Kept
Digital engagement	Digital_Engagement, NumWebVisitsMonth	Distinguishes digitally-engaged customers from low-tech / offline-only customers.
Marketing responsiveness	Total_Campaign_Acceptance, Response, AcceptedCmp1-5	Identifies which customers respond to marketing pushes — useful both for segmentation and for prioritizing future campaigns.
Loyalty / recency	Customer_Tenure_Days, Recency, Customer_Activity_Level_Encoded, Complain	Separates new vs. long-term customers and active vs. lapsing ones; Complain flags at-risk, dissatisfied customers even though it's a small signal.

3. Features I Considered Dropping But Kept Anyway

Product_Preference and Preferred_Shopping_Channel are technically redundant with the underlying Mnt* and Num*Purchases columns (they're derived from them), but I kept both the raw columns and the derived 'winner' category. The raw columns capture magnitude, while the derived category captures a clean, one-hot-encodable signal of which single channel/product dominates — that's useful even though there's some correlation, since clustering algorithms can weigh a clear categorical signal differently than a handful of correlated continuous ones.

4. Summary

Final selected feature count: **40 features** (before scaling) covering demographics, household structure, spending, channel behavior, price sensitivity, digital engagement, marketing responsiveness, and loyalty. After scaling and encoding this becomes 41 numeric columns (25 scaled continuous/ordinal features + 16 one-hot category columns) plus the ID column carried through for traceability. This set is what feeds Task 4 (transformation), Task 5 (scaling), and ultimately the clustering step.